AP Statistics S2 Project Rubrics

AP-style Question Design & Rubric Creation

Weight:10% Group size：1-2 students

Due：2026 Apr 2nd (Thursday)

Design three multiple-choice questions and one free response question in the style of the AP statistics exam, and also develop a scoring guideline.

1. **General Requirements :**
   - A cover page must be included, containing:
     (a) Project name
     (b) Student name (English & Chinese)
     (c) Class number
   - Main text font: Times New Roman, font size: 12 pt
   - Electronic version only (handwritten submissions will not be accepted)
   - The project must be original. AI tools may be used as assistance, but plagiarism is not allowed and will result in an automatic score of zero.
   - Formulas must be created using Microsoft Equation Editor, LaTeX, or MathType. Handwritten formulas are not acceptable.
   - Submit both a Word document and a PDF document through emails: yolandagy@gdhfi.com.

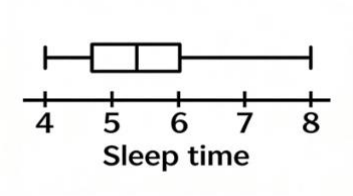
The above requirements must be met; failure to do so may result in point deductions at the instructor's discretion.

2. **Requirements for Multiple-Choice Questions (2% for each, 6% in total):**
   - At least one question must include a statistical graph with clear labels (2%)
   - At least one question must require the use of a calculator (2%)
   - Each question must have five choices (0.5% for each, 1.5% in total)
   - Correct answers and brief explanations must be provided (0.5% for each, 1.5% in total)
   - Scenarios must be clear complete, and realistic (1% for each, 3% in total)

3. **Free Response Question (4%)**
   - Choose one of the following two topics (1%):
     (a) Statistical inference (confidence interval/hypothesis test)
     (b) Probability/Random variable (conditional/binomial/geometric/transforming/combining…)
   - The question must be clearly stated (1%):
     1–2 parts for the chosen topic (a) or 3 parts for the chosen topic (b)
   - Solutions must be clearly explained and aligned with AP exam standards (1%)
   - A scoring guideline must be provided, consistent with AP exam requirements (1%)

MCQ Example

To understand the sleep habits of all 400 HFI students, Yolanda plans to conduct a sample survey. During class on Monday, she asked the 35 students in her statistics class to complete a questionnaire and recorded each individual's sleep time (in hours). The following boxplot summarizes the results.



Yolanda wants to construct a confidence interval to estimate the mean sleep time of all HFI students. Have all conditions for inference been met?

A. Yes, all conditions have been met.

B. No, the distribution of the sample data is not approximately symmetric.

C. No, the sample size is greater than 10 percent of the population size.

D. No, the distribution of sleep time in the population cannot be assumed to be approximately normal.

E. No, the sample was not selected at random.

Solution:

E. This is a convenience sample, it is not random.

FRQ Example

- ✓ 2019-Question 4 (statistical inference)
- ✓ 2018-Question 3 (probability/ random variable)
- ✓ 2018-Question 4 (statistical inference)
- ✓ Other questions given by appendices